SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA10 COURSE CODE: Software Engineering

COURSE OUTCOME COVERED

CO1: *Apply agile methodologies and project management techniques such as scale and lean to effectively manage software projects using industry-standard tools like Jira. (BL1, BL3)*

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks:25**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Concept Mapping

1. Create a concept map that **identifies** and **illustrates** the *Introduction and Brief History of Software Engineering* by linking the following concepts:

- Early software crisis
- Structured development
- Evolution of practices
- Engineering discipline
- Modern systems

Analyze how software engineering evolved over time.

2. Create a concept map that **lists** and **constructs** the *Software Development Life Cycle (SDLC)* by linking the following concepts:

- Requirements
- Design
- Implementation
- Testing
- Maintenance

Explain how each phase is interdependent.

3. Create a concept map that **defines** and **applies** the relationship between *Software Engineering* and *SDLC* by linking the following concepts:

- Process models
- Quality assurance
- Documentation
- Project management
- Lifecycle

Analyze how SDLC supports engineering practices.

4. Create a concept map that **identifies** and **compares** traditional and modern development approaches by linking the following concepts:

- Waterfall
- Agile
- Iteration
- Flexibility
- Feedback

Analyze how methodologies have evolved.

5. Create a concept map that **lists** and **applies** SDLC in project success by linking the following concepts:

- Planning
- Risk management
- Resources
- Time
- Quality

Explain how SDLC ensures successful delivery.

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 – Exemplary (5 Marks)**	**Level 3 – Proficient (4 Marks)**	**Level 2 – Developing (2–3 Marks)**	**Level 1 – Beginning (0–1 Mark)**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand